

Practice problems #7

1. Suppose $x = \begin{bmatrix} 4i \\ 3-i \\ 1 \end{bmatrix}$ is an eigenvector corresponding to the eigenvalue $\lambda = \alpha + i\beta$ of a 3×3 *real* matrix A .
 - (a) Find one more eigenvalue and its corresponding eigenvector *without any calculation*.
 - (b) How many linearly independent eigenvectors in $\mathbf{C}^3$ does A have? Why?
2. For $A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$, find the eigenvalues. Find a basis of $\mathbf{C}^2$ that consists of only eigenvectors of A , if possible.

Answers or Hints on page 2.

#1(a): Since $Ax = \lambda x$, what is $\overline{Ax}$? Note that A is real!

#1(b): 3. How many distinct eigenvalues are possible? Can a complex eigenvalue $\lambda = \alpha + i\beta$ have algebraic multiplicity = 2? Why or why not? (Recall that if λ is an eigenvalue, then $\overline{\lambda}$ is also an eigenvalue. So, the third eigenvalue must be a real number.)

#2: You will get one eigenvalue $2 + i$ with a corresponding eigenvector $\begin{bmatrix} -1 + i \\ 1 \end{bmatrix}$. You can get a second eigenvector easily. Give a basis!